

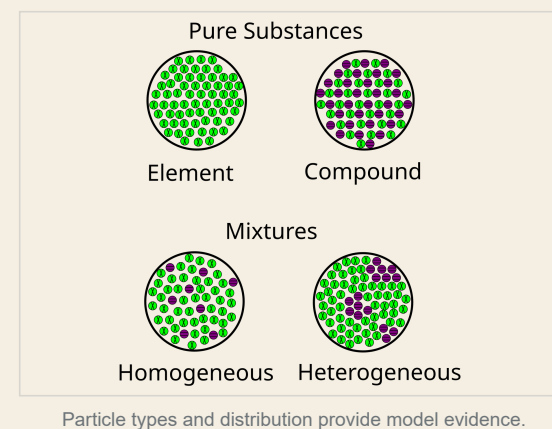
LEARNING TARGET

Use the definition of matter and a particle model to make and limit claims about a sample.

NOTICE / PREDICT

Air is invisible, light is visible, and sound can be heard. Which of the three is matter, and what evidence should decide?

Be ready to defend your first idea. You may revise it after the lesson.



BIG IDEA

Matter has mass and occupies space. Particle models help explain composition and distribution, but their symbols, colors, spacing, and scale are simplified.

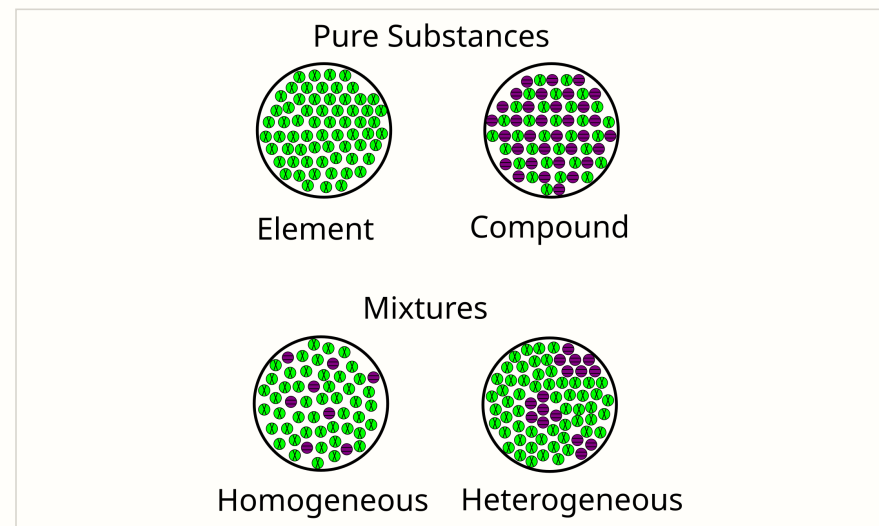
ESSENTIAL TERMS

matter: anything that has mass and occupies space

macroscopic: the scale of observations made directly or with ordinary tools

particle model: a simplified representation of particles and their arrangement

model limitation: a feature of reality that a representation does not show accurately



Particle types and distribution provide model evidence.

WHAT TO REMEMBER

- Use mass and occupied space—not visibility—as the classroom test for matter.
- A particle symbol represents a type of particle; repeated symbols can reveal composition and distribution.
- Identical particles support a pure-substance claim, while more than one particle type supports a mixture claim.
- Models are not to scale and do not prove every property of the real sample.

WORKED EXAMPLE

A sealed diagram contains identical two-atom particles. What can be concluded, and what remains unknown?

1 Every represented particle is identical, so the model supports a pure-substance claim.

2 Inspect the atom symbols inside one particle: the same element would indicate an element; different elements would indicate a compound.

3 If the symbols do not distinguish atom identity, the model lacks the evidence needed for the second decision.

RESULT

The sample is modeled as pure. Element versus compound cannot be decided without evidence about the atom types in each particle.

CHECK: Why does visibility not determine whether something is matter?

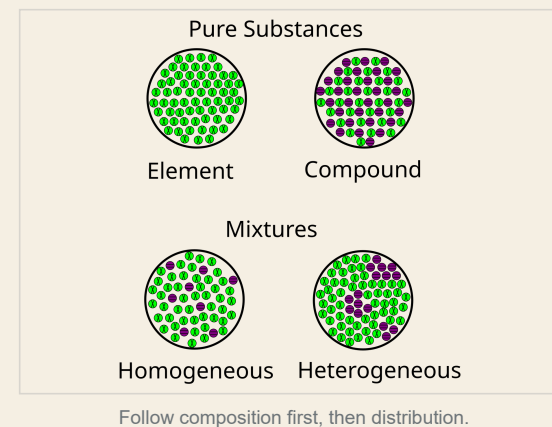
LEARNING TARGET

Classify elements, compounds, homogeneous mixtures, and heterogeneous mixtures from composition evidence.

NOTICE / PREDICT

**A clear saltwater sample looks uniform.
Does uniform appearance prove that it is
a pure substance?**

Be ready to defend your first idea. You may revise it after the lesson.



BIG IDEA

Classification depends on how many substances are present, whether composition is fixed, and whether multiple components are uniformly distributed.

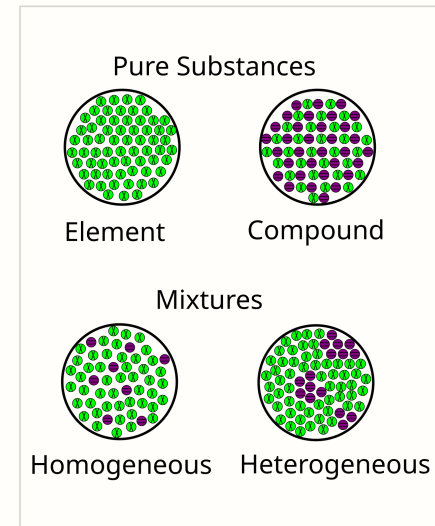
ESSENTIAL TERMS

pure substance: matter with constant composition and characteristic properties

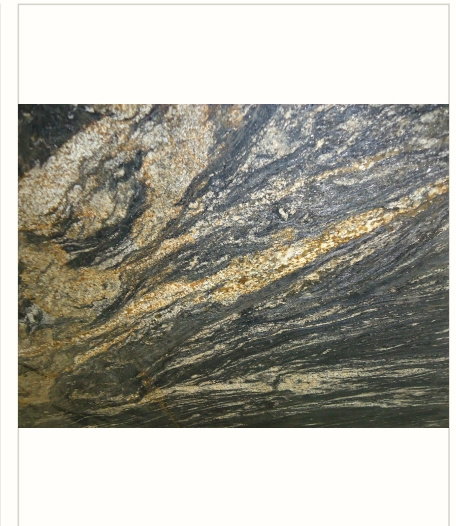
element: a pure substance made of one chemical element

compound: a pure substance containing elements chemically combined in fixed proportions

mixture: a physical combination of substances with variable composition



Follow composition first, then distribution.



Visible mineral regions make granite nonuniform at this scale.

WHAT TO REMEMBER

- First decide pure substance versus mixture; then choose the more specific label.
- An element contains one element type; a compound contains different elements chemically combined in a fixed composition.
- A homogeneous mixture is uniform at the sampled scale but still contains multiple substances.
- A heterogeneous mixture has nonuniform composition or visibly distinct regions. Physical separation evidence supports a mixture claim.

WORKED EXAMPLE

A clear saltwater sample leaves salt after the water is removed by a physical process. Classify the original sample.

1

The sample appears uniform, so it is homogeneous at the observed scale.

2

A physical process recovers components, which supports the presence of more than one substance.

3

Uniform distribution plus multiple substances identifies a homogeneous mixture.

RESULT

Saltwater is a homogeneous mixture, not a pure compound.

CHECK: Why are the words homogeneous and pure not interchangeable?

LEARNING TARGET

Classify physical and chemical properties and distinguish intensive from extensive properties.

NOTICE / PREDICT

Copper conducts electricity and can react with oxygen. Which statement can be checked without changing copper into a new substance?

Be ready to defend your first idea. You may revise it after the lesson.



Color and luster are observable physical properties of copper.

BIG IDEA

A physical property can be observed or measured without changing chemical identity; a chemical property describes the ability to undergo a chemical change.

ESSENTIAL TERMS

physical property: a characteristic observed or measured without changing identity

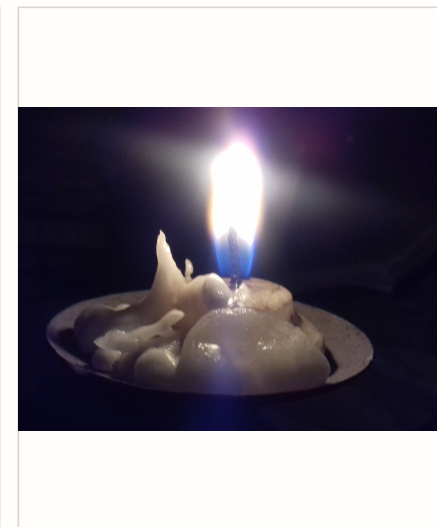
chemical property: the ability of a substance to undergo a chemical change

intensive property: a property independent of sample amount under stated conditions

extensive property: a property that depends on the amount of sample



Color and luster are observable physical properties of copper.



Flammability describes the ability to burn; the property exists before ignition.

WHAT TO REMEMBER

- Color, density, melting point, and conductivity are physical properties when measured without changing identity.
- Flammability and reactivity describe possible chemical behavior, even when no reaction is occurring at that moment.
- Mass and volume are extensive because they change with sample amount.
- Density and melting point are intensive under the same stated conditions and can help identify a substance.

WORKED EXAMPLE

Classify density and mass as physical or chemical and as intensive or extensive.

1 Both can be measured without changing chemical identity, so both are physical properties.

2 Density remains characteristic when sample amount changes under the same conditions, so it is intensive.

3 Mass increases or decreases with sample amount, so it is extensive.

RESULT

Density is physical and intensive; mass is physical and extensive.

CHECK: Why can flammability be stated as a property before a flame is present?

LEARNING TARGET

Classify changes by the new-substance criterion and evaluate common clues in context.

NOTICE / PREDICT

Bubbles appear when a cold soda is opened and when two reacting solutions are mixed. Do bubbles prove the same kind of change in both cases?

Be ready to defend your first idea. You may revise it after the lesson.



Melting changes state while the substance remains water.

BIG IDEA

A chemical change forms one or more new substances. Color, gas, precipitate, light, odor, or temperature change can support a claim but is not automatic proof.

ESSENTIAL TERMS

physical change: a change in form or state that does not require a new substance

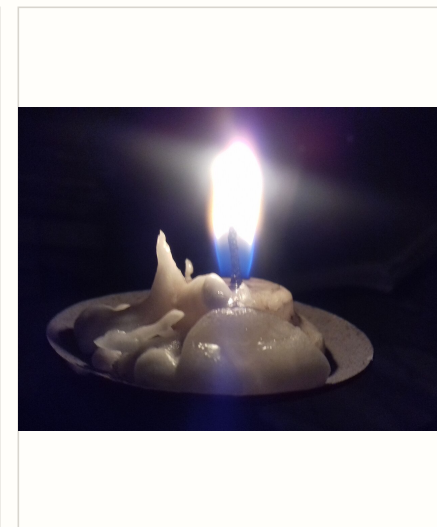
chemical change: a process in which atoms rearrange to form new substances

evidence: an observation or result used to support or challenge a claim

conservation of matter: matter is not created or destroyed in a closed-system chemical change



Melting changes state while the substance remains water.



Burning forms new substances and is chemical.

WHAT TO REMEMBER

- Melting, freezing, crushing, dissolving in a recoverable context, and physical separation do not by themselves require a new substance.
- Burning, decomposition, and reactions that form products with new compositions are chemical changes.
- A clue must be compared with alternatives: bubbles may be a gas product, boiling, or released dissolved gas.
- An open container may lose gaseous products from measurement; this does not violate conservation of matter.

WORKED EXAMPLE

Salt dissolves in water and is later recovered by evaporating the water. Classify the process in this context.

- 1 Dissolving produces a uniform solution but does not by itself establish a new substance.
- 2 The original salt is recovered by a physical method.
- 3 Recovery supports a physical-change/separation claim for this classroom context.

RESULT

Physical change and physical separation in this context.

CHECK: Name one nonchemical process that can produce bubbles.

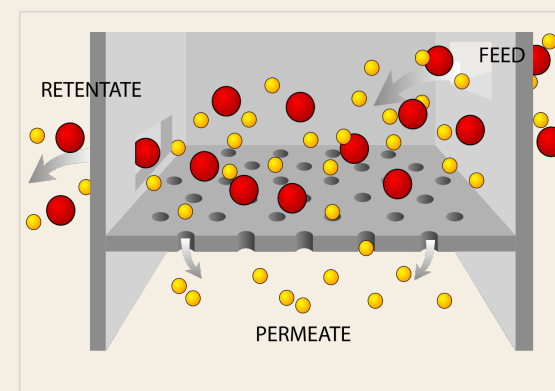
LEARNING TARGET

Choose and sequence physical separation methods from differences in particle size, solubility, volatility, magnetism, or phase behavior.

NOTICE / PREDICT

A beaker contains iron filings, salt, sand, and water. Which separation should happen first, and why does order matter?

Be ready to defend your first idea. You may revise it after the lesson.



Residue remains on the filter; filtrate passes through.

BIG IDEA

A separation method works because components differ in a physical property. A good plan names both the method and the property difference.

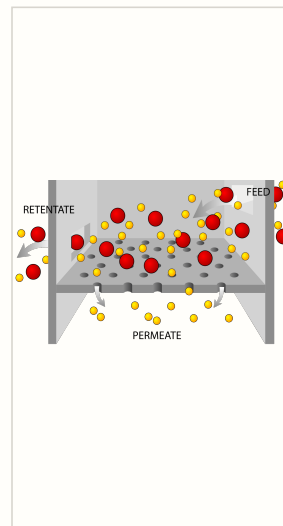
ESSENTIAL TERMS

filtration: separation using particle size and insolubility

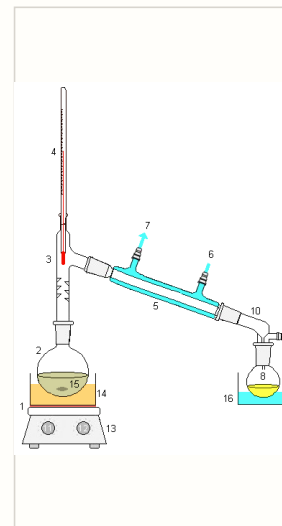
distillation: separation using differences in volatility or boiling behavior

chromatography: separation using different attractions and travel rates

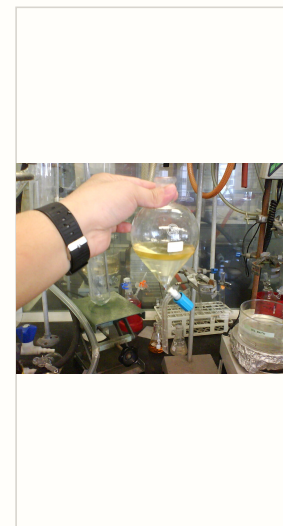
decantation: separation by carefully pouring one phase from another



Residue remains on the filter; filtrate passes through.



Vapor is condensed and collected during simple distillation.



Immiscible liquid layers can be drained separately.

WHAT TO REMEMBER

- Use magnetism before adding water when a dry magnetic component is present.
- Use filtration for an insoluble solid in a liquid; dissolved solute passes with the liquid.
- Use distillation when the goal includes collecting the more volatile liquid as well as the residue.
- Use chromatography to test whether a soluble colored sample contains components with different attractions and travel rates.

WORKED EXAMPLE

A uniform blue liquid yields colorless liquid and blue solid after distillation. What claim is supported?

- 1 The original sample appears homogeneous, but appearance alone cannot establish purity.
- 2 Distillation is a physical method based on volatility or boiling behavior.
- 3 Recovering two components by a physical process supports a mixture claim.

RESULT

The original sample is a homogeneous mixture with at least two components.

CHECK: Why does filtration fail to remove salt dissolved in water?